

$\underline{X} = \begin{pmatrix} X_1 \\ \vdots \\ X_k \end{pmatrix}$ random vector of dimension k

$$E\underline{X} = \underline{\mu} \quad \text{Var}(\underline{X}) = \underline{\Sigma} = (\sigma_{ij})$$

$$= \begin{pmatrix} E X_1 \\ \vdots \\ E X_k \end{pmatrix} = \begin{pmatrix} \mu_1 \\ \vdots \\ \mu_k \end{pmatrix}$$

$$\sigma_{ij} = \text{cov}(X_i, X_j) \quad i, j = 1, \dots, k$$

$$\text{cov}(X_i, X_i) = \text{Var}(X_i).$$

$A_{m \times k}$ constant matrix, $\underline{b}_m$ constant vector.

$$Y = AX + b \quad m\text{-dim random vector.} \quad \boxed{EY = A\underline{\mu} + b}$$

Want EY and $\text{Var}(Y)$.

→ When $k=1$ and $m=1$.

$$EY = A\underline{\mu} + b \quad \text{Var } Y = A^2 \text{Var}(X)$$

with scalars
 $E(ax+by) = aE(x) + bE(y)$

$$\textcircled{1} \quad Y = \begin{pmatrix} Y_1 \\ \vdots \\ Y_m \end{pmatrix}$$

$$Y_i = \sum_{j=1}^k a_{ij} X_j + b_i$$

$$A = (a_{ij}) \quad \underline{b} = \begin{pmatrix} b_1 \\ \vdots \\ b_m \end{pmatrix}$$

$$EY_i = \sum_j a_{ij} EX_j + b_i = \sum_j a_{ij} \mu_j + b_i \quad i=1, \dots, m$$

$$EY = \begin{pmatrix} EY_1 \\ \vdots \\ EY_m \end{pmatrix} = A \begin{pmatrix} EX_1 \\ \vdots \\ EX_k \end{pmatrix} + b = A\underline{\mu} + b.$$

Fix i $\text{Var}(Y_i) = \text{Var}\left(\sum_j a_{ij} X_j + b_i\right) = \text{Var}\left(\sum_j a_{ij} X_j\right)$

$$= \sum_j a_{ij}^2 \text{Var}(X_j) + \sum_{j \neq k} a_{ij} a_{ik} \text{cov}(X_j, X_k)$$

$$= (a_{i1} \dots a_{ik}) \underline{\Sigma} \begin{pmatrix} a_{i1} \\ \vdots \\ a_{ik} \end{pmatrix}$$

$$\text{cov}(Y_i, Y_l) = \text{cov}\left(\sum_j a_{ij} X_j + b_i, \sum_{kn} a_{ln} X_{kn} + b_l\right)$$

$$= \sum_j \sum_{kn} a_{ij} a_{ln} \text{cov}(X_j, X_{kn})$$

$$= (a_{i1} \dots a_{ik}) \underline{\Sigma} \begin{pmatrix} a_{l1} \\ \vdots \\ a_{lk} \end{pmatrix}$$

$$\textcircled{2} \quad \text{Var}(Y) = A \underline{\Sigma} A^T$$

with scalars
 $\text{Var}(ax+by)$
 $= a^2 \text{Var}(x) + b^2 \text{Var}(y)$
 $+ 2ab \text{cov}(x, y)$
 $+ ab \text{cov}(y, x)$

$$\underline{\Sigma} = (\sigma_{ij})$$

$$\underline{\text{cov}}(ax+by, cx+dy)$$

$$= ac \text{Var}(x) + bd \text{Var}(y)$$

$$+ ad \text{cov}(x, y) + bc \text{cov}(x, y)$$

Model → $Y = X\beta + \varepsilon$ $E(\varepsilon) = 0$ $\text{Var}(\varepsilon^2 I)$

dim → β_{p+1} Y_n $X_{n \times (p+1)}$ ε_n

LS ~~opt~~ criterion $\rightarrow y^T y - 2 y^T X \beta + \beta^T (X^T X) \beta = \varepsilon^T \varepsilon$

LS estimate $\rightarrow \hat{\beta} = (X^T X)^{-1} X^T y$

$E\hat{\beta} = \beta$ $\text{Var}(\hat{\beta}) = \sigma^2 (X^T X)^{-1}$

Data analysis : Diagnostics

ANOVA.

Source	SS	df	MS	F
Regression	SS _{Reg}	p		
Residual	SS _{Res}	n - (p+1)		
Total	SST	n - 1		

~~$E(SS) = \sigma^2$~~
 $E(SS/\sigma^2) = df$

$SST = \sum_{i=1}^n (y_i - \bar{y})^2 = y^T y - y^T 1 1^T y$

$SS_{Res} = \sum (y_i - \hat{y}_i)^2 = (y - \hat{y})^T (y - \hat{y})$
 $= (y - X\hat{\beta})^T (y - X\hat{\beta})$
 $= y^T y - 2 \hat{\beta}^T X^T y + \hat{\beta}^T X^T X \hat{\beta}$
 $= y^T y - y^T X (X^T X)^{-1} X^T y$

$\hat{y} = X\hat{\beta}$
 $\hat{\beta} = (X^T X)^{-1} X^T y$

$SS_{Reg} = SST - SS_{Res}$

Model: $y_i = \beta_0 + \beta_1 x_{1i} + \dots + \beta_p x_{pi} + \varepsilon_i, i=1, \dots, n$

Data: $(x_{1i}, \dots, x_{pi}, y_i), i=1, \dots, n$

Matrix notation $y = X\beta + \varepsilon$

p+2 parameters
 $\beta_0, \beta_1, \dots, \beta_p, \sigma^2$

Assumptions: $E\varepsilon_i = 0, \text{Var} \varepsilon_i = \sigma^2 \forall i$

Least Squares criterion.

Least squares estimates of β : $\hat{\beta} = (X^T X)^{-1} X^T y$

Expectation & Var $E\hat{\beta} = \beta, \text{Var}(\hat{\beta}) = \sigma^2 (X^T X)^{-1}$

Sum of Squares & ANOVA

$SS_{Res} = y^T y - y^T [X(X^T X)^{-1} X^T] y$ df: n - (p+1)

$SST = y^T y - y^T 1 1^T y$ df: n - 1

$SS_{Reg} = SST - SS_{Res}$ df: p

Diagnostic plots: ε vs $\hat{y}$, $|\hat{\varepsilon}|$ vs $\hat{y}$, QQ plot, outliers, normality

Diagnostic plots : $\hat{e}$ vs $\hat{y}$, $|\hat{e}|$ vs $\hat{y}$, $Q-Q$ plot, outliers
 linear model σ^2 is const. normality

Example with data.

Estimate of σ^2 : $\text{Var}(E_i) = \frac{1}{n-1} \sum_{i=1}^n (y_i - \bar{y})^2$

$$\frac{1}{n-1} \sum (\hat{e}_i - \bar{\hat{e}})^2 \quad \hat{e}_i = y_i - \hat{y}_i$$

Claim: $\bar{\hat{e}} = 0$.

Claim 2: $E(\sum \hat{e}_i^2) = (n-p+1)\sigma^2$ [Not proved]

$$\hat{\sigma}^2 = \frac{1}{n-p+1} \sum \hat{e}_i^2 = \frac{SS_{Res}}{n-p+1} \text{ unbiased}$$

Claim 1: $\hat{y} = X\hat{\beta} = X(X^T X)^{-1} X^T y$ [not proved]

$p=1$. $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i = \bar{y} - \hat{\beta}_1 \bar{x} + \hat{\beta}_1 x_i$

$$\sum \hat{e}_i = \sum (y_i - \hat{y}_i) = \sum (y_i - \bar{y} + \hat{\beta}_1 \bar{x} - \hat{\beta}_1 x_i) = n\bar{y} - n\bar{y} + \hat{\beta}_1 n\bar{x} - \hat{\beta}_1 n\bar{x} = 0$$

$$R^2 = \frac{SS_{\text{explained}}}{SS_{\text{total}}} = 1 - \frac{SS_{\text{unexplained}}}{SS_{\text{total}}} = 1 - \frac{SS_{Res}}{SST} = \frac{SS_{Reg}}{SST}$$

Good to have high R^2 .

Claim: Keep adding more covariates This will ~~decrease~~ ^{increase} R^2 .
 $\beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 \log(x_4/2) + \dots$

More parameters are not necessarily good. Overfitting.
 Very unstable estimates
 Prediction for new datapoints bad.

Some compromise for the increase in number of parameters

Adjusted R^2 is obtained after dividing by df.

Ratio of MS and not SS.

Pf of claim. To show SS_{Res} decreases if we add a covariate.

(A) $SS_{Res}_p = \min_{\beta} \sum_{i=1}^n (y_i - \sum_{j=1}^p \beta_j x_{ji} - \beta_0)^2 = \min_{\beta_0, \dots, \beta_p} \sum_{i=1}^n (y_i - \sum_{j=1}^p \beta_j x_{ji} - \beta_0 - 0 \cdot x_{pi})^2$

(B) $SS_{Res}_{p+1} = \min_{\beta} \sum_{i=1}^n (y_i - \sum_{j=1}^{p+1} \beta_j x_{ji} - \beta_0)^2 = \min_{\beta} \sum_{i=1}^n (y_i - \sum_{j=1}^p \beta_j x_{ji} - \beta_0 - \beta_{p+1} x_{p+1,i})^2$

$$b) SSR_{\beta_{p+1}} = \min_{\beta} \left(y_i - \sum_{j=1}^{p+1} \beta_j x_{ji} - \beta_0 \right)^2 = n \sum_{i=1}^n \left(y_i - \sum_{j=1}^p \beta_j x_{ji} - \beta_0 - \beta_{p+1} x_{p+1,i} \right)^2$$

$$SSR_{\beta_{p+1}} < SSR_{\beta} \quad (A) \text{ is a constrained version of } (B)$$

Linear model

- ① Linear vs. non-linear.
↓
in parameters.

Linear $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2^2 + \beta_3 \ln x_1 + \epsilon$

$y = \frac{\beta_0 + \beta_1 x_1}{\beta_2 + \beta_3 x_2}$ non linear.

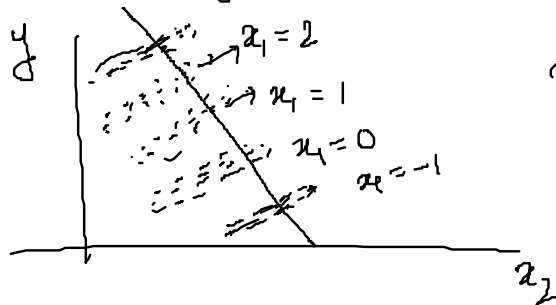
- ② Correlation is not causation.

Correlated variables will give a good linear fit.

- ③ Interpretation of coefficients.

$$y = \beta_0 + \beta_1 x_1 + \boxed{\beta_2} x_2 + \dots + \beta_p x_p + \epsilon.$$

β_2 : change in y for unit change in x_2 , when all other x_j 's are held at the same level.



Regression of y on x_2 gives $\beta_2 < 0$

Condition on each value of x_1
then $\beta_1 > 0$.

Simpson's Paradox.